

Toward Fair Neuroimaging AI

From dementia classification to brain age prediction

Core question

Can neuroimaging AI stay accurate and fair across populations?

1 • Dementia classification

2 • Brain age prediction



Ngoc Huynh Ho • Neuroimage Analytics Laboratory • UT Health Science Center at San Antonio

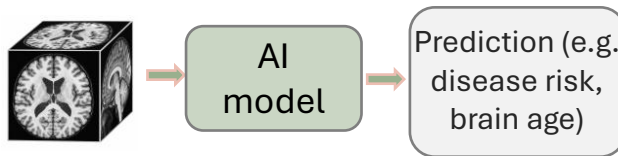
10/07/2026

High accuracy does not guarantee equitable performance

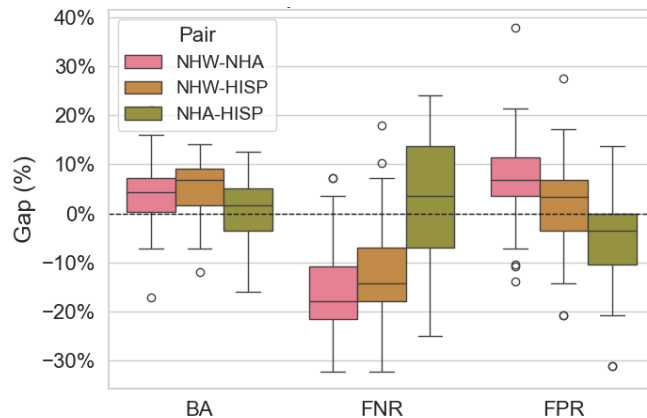
Overall metrics can hide failing performance in specific populations.

Conventional model selection

Optimize only overall performance

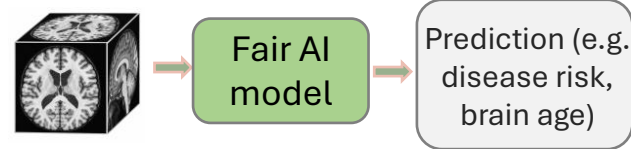


Good average accuracy but larger subgroup gaps

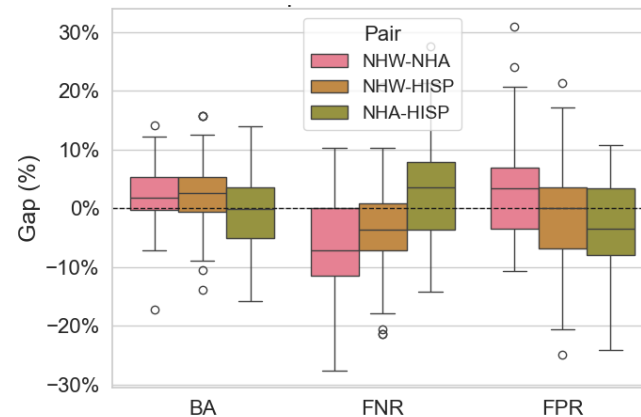


Fairness-aware model selection

Optimize only overall performance
+ fairness



More consistent performance and
reduce subgroup gaps



Clinical relevance

False negatives may delay dementia care.
False positives may create unnecessary burden.

Guiding question

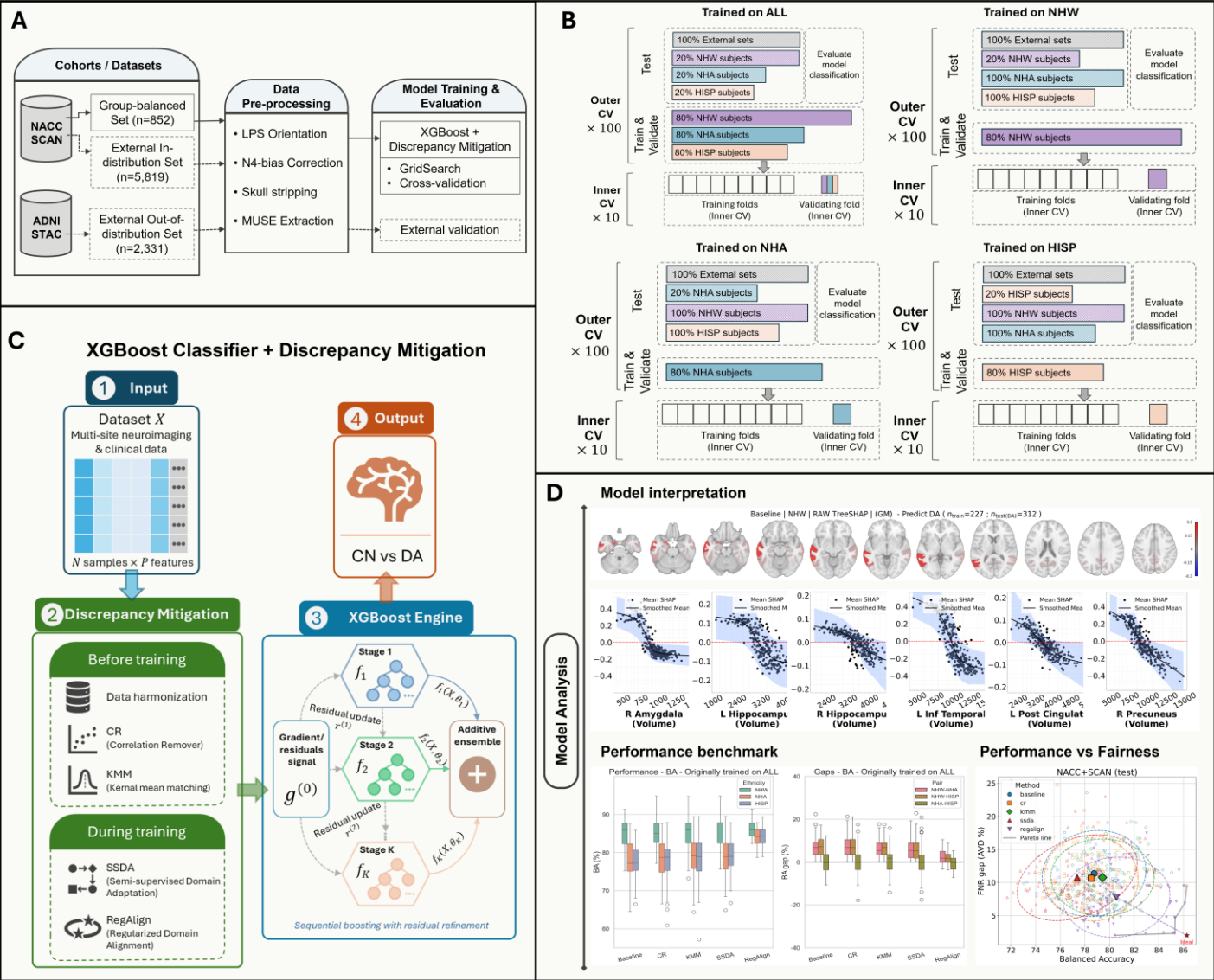
Do disparities remain after controlling for sample size, demographics, and imaging-site effects?

Framing

Fairness is not only a social concern — it is a reliability problem for clinical AI.

Research topic 1: Advancing fair and explainable machine learning for neuroimaging dementia pattern classification in multi-racial and multi-ethnic populations

Identify, mitigate, and interpret racial/ethnic performance disparities in MRI-based dementia classification.



Multi-cohort evaluation

NACC + SCAN for model development and fairness analysis; ADNI + STAC for external validation.

Fairness assessment

Compare performance across NHW, NHA, and HISP populations using balanced accuracy, false-negative rate, and false-positive rate.

Discrepancy mitigation

Evaluate data-level and learning-based strategies, including fairness-aware adaptation, to reduce inter-group performance gaps.

Model interpretation

Use TreeSHAP and regional dependence analysis to identify population-specific neuroimaging patterns associated with model predictions.

Fairness-aware dementia classification

MRI-based dementia classification across multi-racial and multi-ethnic cohorts.

Variable	Overall (n = 9007)	NHW (n = 6584)	NHA (n = 1263)	HISP (n = 713)	Other (n = 447)	Test Statistic (p)
NACC / SCAN						
Diagnosis (CN/DA)	4834/1842	3487/1490	839/147	311/142	197/63	$\chi^2(3) = 97.77, p = 4.70e^{-21}$ $V = 0.121, 95\% \text{ CI } [0.103, 0.142]$
Age (years)						
CN	70.5 ± 10.0	70.5 ± 10.5	70.6 ± 8.0	70.5 ± 9.9	68.8 ± 8.4	$F(3, 4830) = 1.89, p = 0.13$ $\eta^2 = 0.001, 95\% \text{ CI } [0.000, 0.004]$
DA	73.0 ± 9.9	72.7 ± 9.8	75.3 ± 9.1	72.9 ± 11.8	74.9 ± 9.3	$F(3, 1838) = 4.07, p = 6.84e^{-3}$ $\eta^2 = 0.007, 95\% \text{ CI } [0.002, 0.016]$
Female, %	62%	58%	75%	67%	67%	$\chi^2(3) = 106.81, p = 5.32e^{-23}$ $V = 0.126, 95\% \text{ CI } [0.106, 0.149]$
Education (years)						
CN	16.2 ± 4.0	16.6 ± 3.1	15.4 ± 3.9	14.0 ± 6.5	16.2 ± 9.1	$F(3, 4830) = 55.52, p = 2.81e^{-35}$ $\eta^2 = 0.033, 95\% \text{ CI } [0.017, 0.063]$
DA	15.3 ± 5.2	15.7 ± 3.6	15.2 ± 10.3	11.8 ± 5.1	14.5 ± 12.1	$F(3, 1838) = 25.75, p = 2.85e^{-16}$ $\eta^2 = 0.040, 95\% \text{ CI } [0.023, 0.101]$
MMSE						
CN	28.7 ± 1.6	29.0 ± 1.3	27.9 ± 2.1	27.9 ± 1.9	28.1 ± 2.0	$F(3, 1045) = 36.28, p = 2.64e^{-22}$ $\eta^2 = 0.094, 95\% \text{ CI } [0.061, 0.140]$
DA	20.4 ± 5.9	21.0 ± 5.6	18.2 ± 6.7	18.3 ± 6.6	18.4 ± 5.4	$F(3, 882) = 11.33, p = 2.69e^{-7}$ $\eta^2 = 0.037, 95\% \text{ CI } [0.018, 0.070]$
ADNI / STAC						
Diagnosis (CN/DA)	1389/942	860/747	231/46	185/75	113/74	$\chi^2(3) = 104.29, p = 1.86e^{-22}$ $V = 0.212, 95\% \text{ CI } [0.180, 0.248]$
Age (years)						
CN	70.5 ± 7.4	71.8 ± 6.8	67.6 ± 7.3	69.1 ± 8.2	68.9 ± 7.6	$F(3, 1385) = 25.86, p = 2.81e^{-16}$ $\eta^2 = 0.053, 95\% \text{ CI } [0.034, 0.080]$
DA	74.5 ± 7.6	74.6 ± 7.6	70.8 ± 7.1	74.5 ± 8.4	75.9 ± 6.5	$F(3, 938) = 4.61, p = 3.29e^{-3}$ $\eta^2 = 0.015, 95\% \text{ CI } [0.005, 0.035]$
Female, %	56%	51%	73%	67%	56%	$\chi^2(3) = 61.71, p = 2.53e^{-13}$ $V = 0.163, 95\% \text{ CI } [0.126, 0.202]$
Education (years)						
CN	16.4 ± 3.4	16.6 ± 2.5	15.9 ± 2.6	15.8 ± 6.9	17.0 ± 2.2	$F(3, 1384) = 5.53, p = 9.03e^{-4}$ $\eta^2 = 0.012, 95\% \text{ CI } [0.003, 0.052]$
DA	15.4 ± 3.0	15.7 ± 2.7	14.2 ± 3.1	13.4 ± 3.5	15.0 ± 4.1	$F(3, 933) = 16.37, p = 2.25e^{-10}$ $\eta^2 = 0.050, 95\% \text{ CI } [0.027, 0.090]$
MMSE						
CN	29.0 ± 1.3	29.1 ± 1.2	28.7 ± 1.3	29.0 ± 1.3	28.8 ± 1.3	$F(3, 1216) = 4.42, p = 4.22e^{-3}$ $\eta^2 = 0.011, 95\% \text{ CI } [0.003, 0.029]$
DA	20.8 ± 5.1	20.9 ± 5.2	21.2 ± 4.0	21.0 ± 4.7	20.2 ± 5.3	$\eta^2 = 0.002, 95\% \text{ CI } [0.000, 0.015]$

Study scope

9,007

participants

3

primary groups

4

cohort sources

Primary groups

Non-Hispanic White (NHW)

Non-Hispanic African American (NHA)

Hispanic (HISP)

Evaluation targets

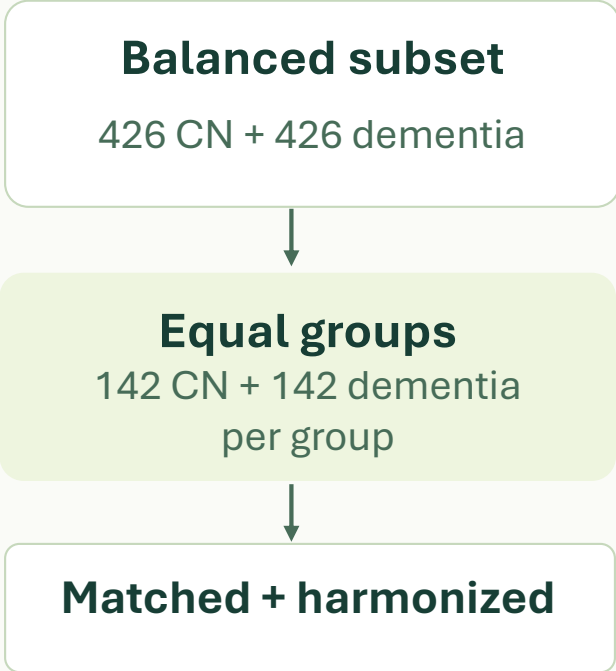
Balanced accuracy, false-negative rate, false-positive rate, and pairwise group gaps.

Ho, NH., Charisis, S., Honnorat, N. et al. Advancing fair and explainable machine learning for neuroimaging dementia pattern classification in multi-racial and multi-ethnic populations. Nat Commun 17, 8026 (June 2026). <https://doi.org/10.1038/s41467-026-74515-w>

Controlled test: remove the obvious confounders first

A strictly balanced subset isolates group-associated model behavior.

Variable	Overall	NHW	NHA	HISP	Test
Diagnosis (CN/DA)	426/426	142/142	142/142	142/142	$\chi^2(2) = 0.00, p = 1$ $V = 0.00, 95\% \text{ CI } [0.01, 0.10]$
AGE (years)					
CN	72.95 ± 6.63	72.92 ± 6.98	73.39 ± 5.68	72.53 ± 7.15	$F(2, 423) = 0.59, p = 0.55$ $\eta^2 = 0.00, 95\% \text{ CI } [0.00, 0.03]$
DA	73.57 ± 10.96	72.88 ± 11.78	74.95 ± 9.05	72.88 ± 11.76	$F(2, 423) = 1.68, p = 0.19$ $\eta^2 = 0.01, 95\% \text{ CI } [0.00, 0.03]$
t-test	$t(700) = -1.00$	$t(229) = 0.03$	$t(237) = -1.74$	$t(233) = -0.30$	
p	0.32	0.97	0.08	0.76	
d	-0.07	0.00	-0.21	-0.04	
95% CI	[-0.21, 0.07]	[-0.23, 0.24]	[-0.45, 0.02]	[-0.28, 0.20]	
Female, %					
CN	64.08%	64.08%	64.08%	64.08%	$\chi^2(2) = 0.00, p = 1$ $V = 0.00, 95\% \text{ CI } [0.01, 0.13]$
DA	64.08%	64.08%	64.08%	64.08%	$\chi^2(2) = 0.00, p = 1$ $V = 0.00, 95\% \text{ CI } [0.01, 0.13]$
Fisher's exact	1	1	1	1	
p	1	1	1	1	
Odds ratio	[0.76, 1.32]	[0.62, 1.62]	[0.62, 1.62]	[0.62, 1.62]	
95% CI	[0.76, 1.32]	[0.62, 1.62]	[0.62, 1.62]	[0.62, 1.62]	

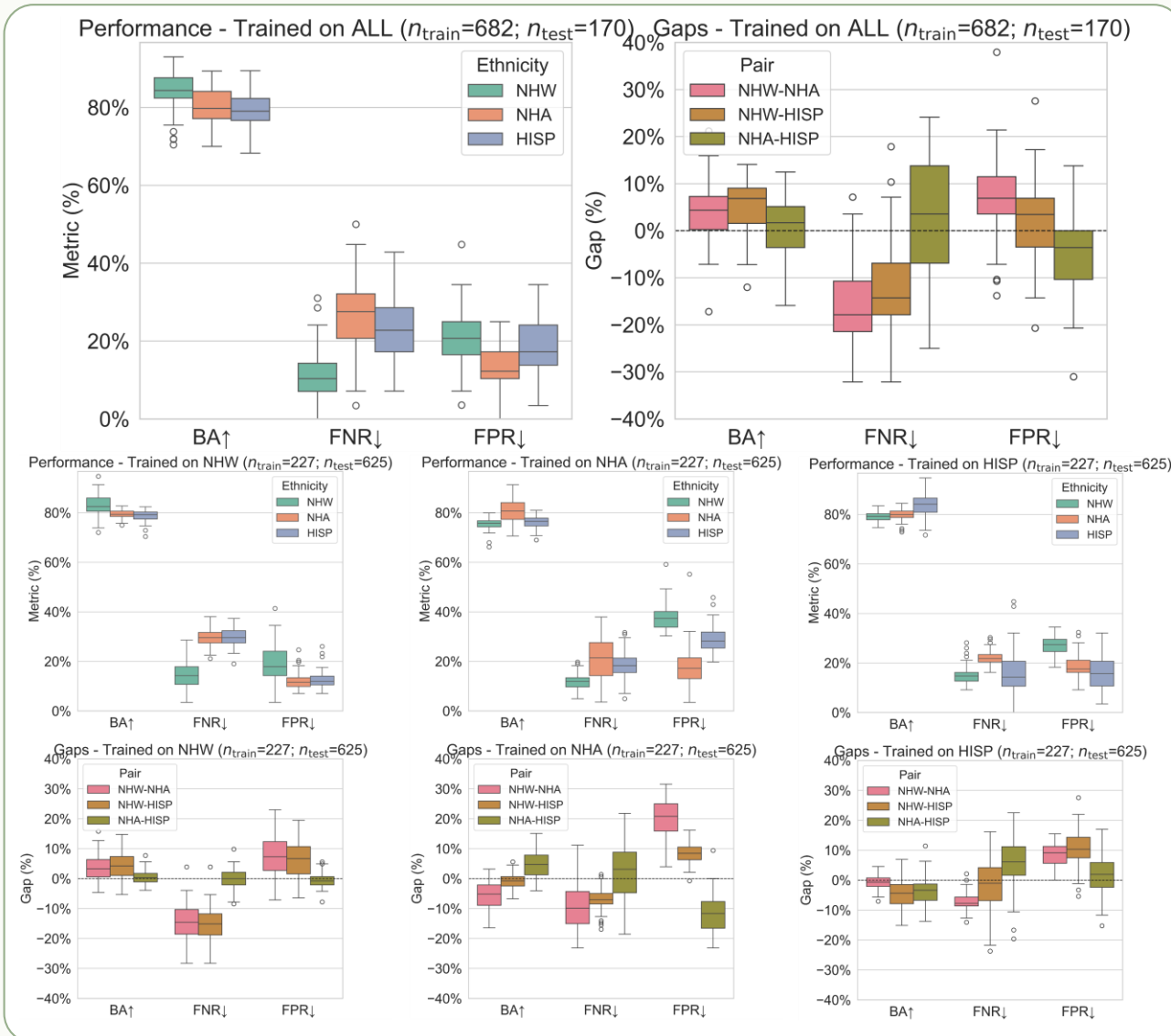


Why this design matters

If disparities persist here, they cannot be dismissed as only unequal sample size or age/sex mismatch.

Population disparities persist under strict balancing

The gap remained even after controlled evaluation.



Main result

Cross-group performance gaps remained across all training scenarios.

14–15%

FNR gap after
NHW-only training

13–16%

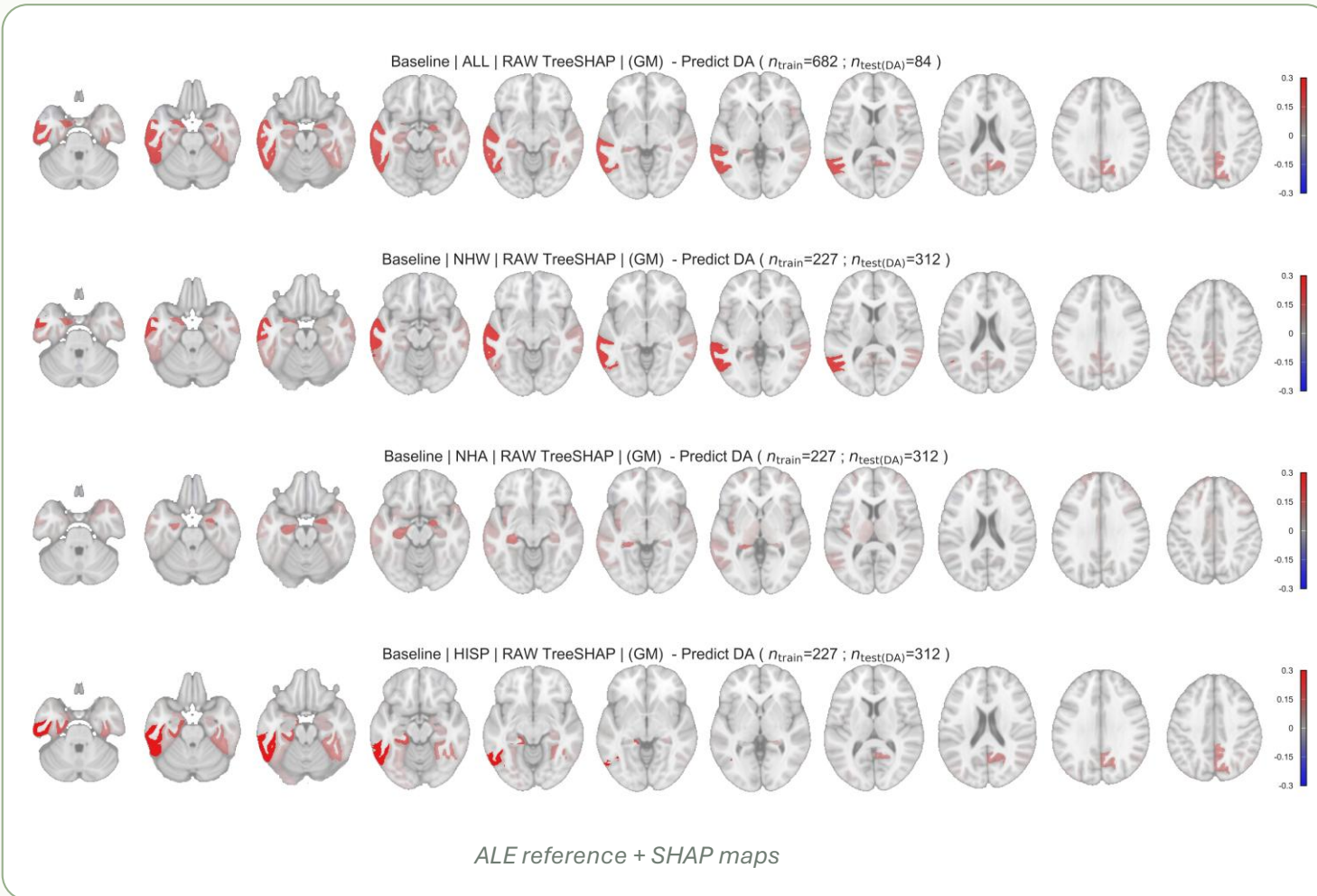
FNR gap even with
ALL groups

Take-home line

Sample size matters — but it does not fully solve fairness.

Fairness differences are visible in what the model learns

Population-specific attribution patterns help interpret model behavior.



Interpretation

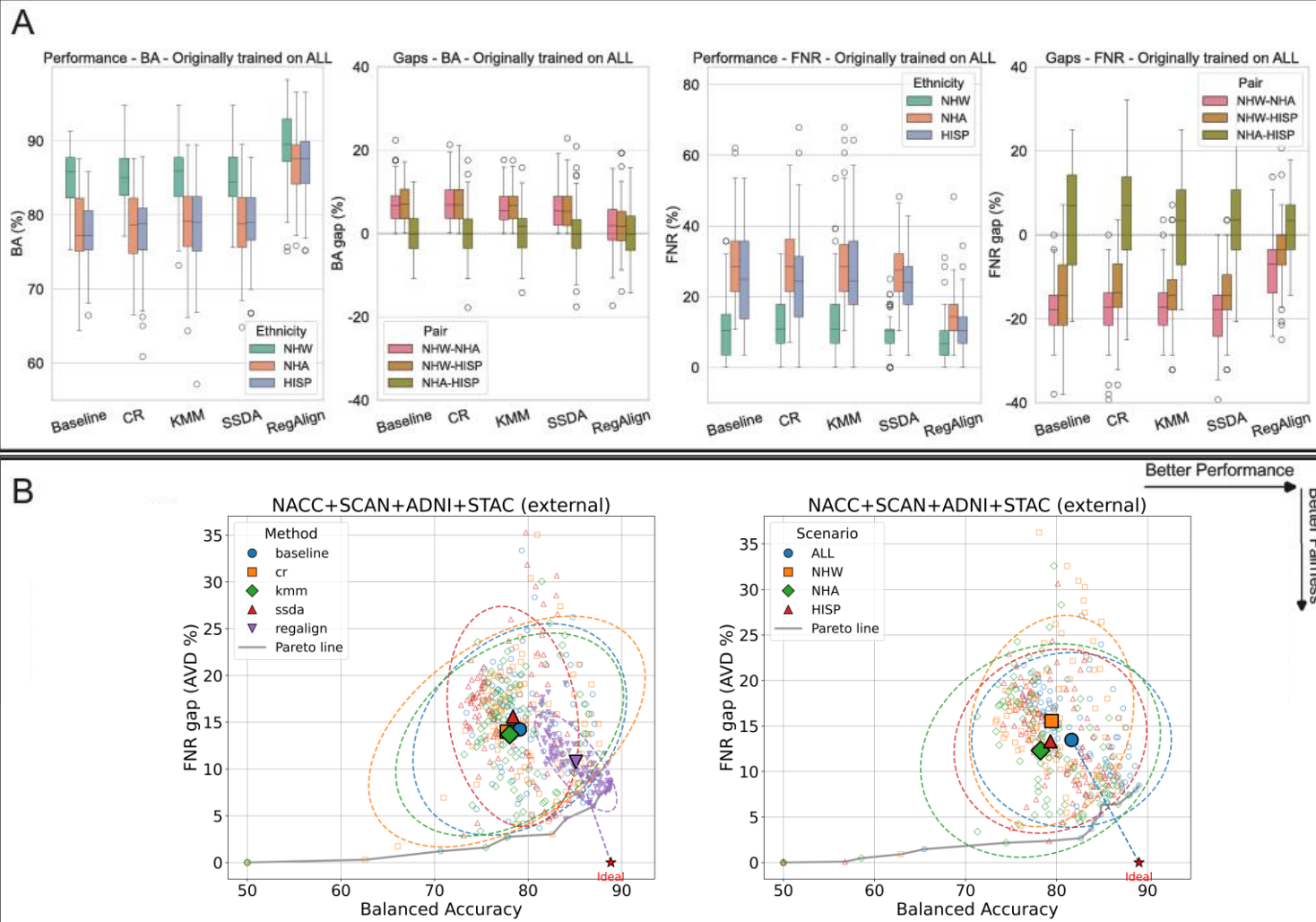
Models overlap with canonical dementia regions, but the relative importance and spatial distribution vary by population.

Regions to mention

Hippocampus • Amygdala
Entorhinal cortex
Parahippocampal gyrus
Precuneus

Fairness-aware adaptation reduces inter-group gaps

Fairness can be an explicit objective, not only a post-hoc audit.



Result focus

Compare whether each strategy narrows discrepancy while preserving useful performance.

Key message

Fairness-aware adaptation can narrow population performance gaps.

Accuracy

Fairness gaps

Reliability

From diagnosis to biomarkers

Brain age reframes fairness as a regression and calibration problem.

Dementia classification

Binary outcome
CN vs dementia
Group gaps in BA / FNR / FPR



Brain age prediction

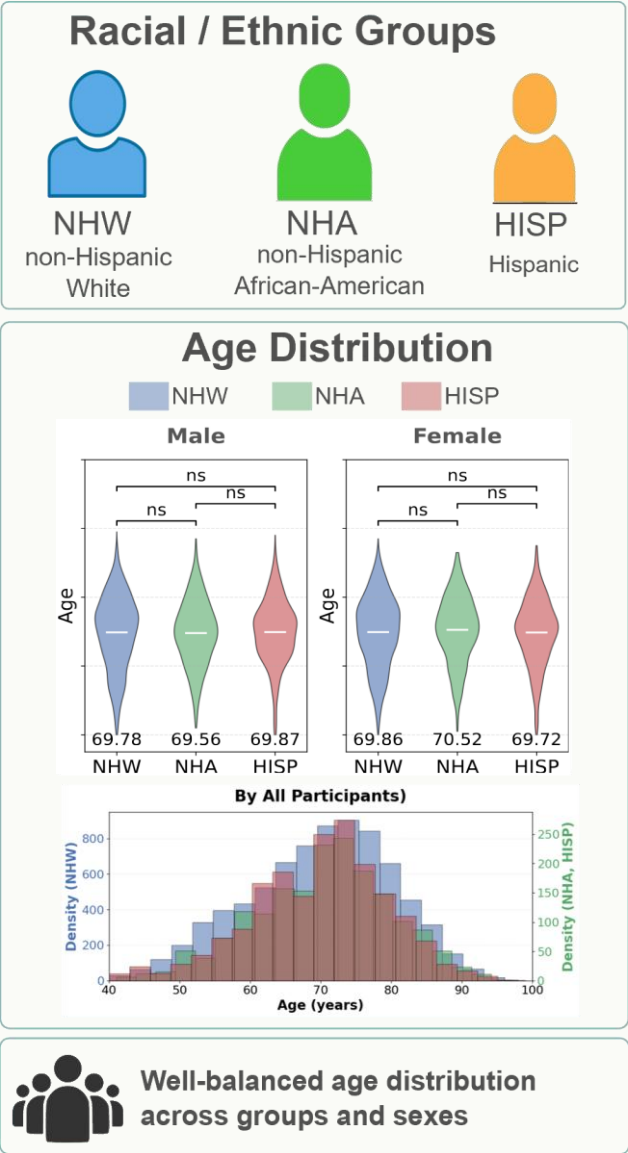
Continuous outcome
Predicted age vs chronological age
Group gaps + age-distribution bias

Key transition

If brain age is used as a biomarker, a low global error is not enough — calibration must be consistent across people and ages.

Research Topic 2: Mitigating racial and age-distribution bias in MRI-based brain age prediction across multi-ethnic cohorts

Large-scale multi-ethnic normative brain-aging model.



Study scope

12,008

T1-weighted MRI scans

13

cohorts

40–99

years

Development + validation

11,453 scans from 12 development cohorts; DVCID external validation, n = 1,352

Results focus

Show that fairness-aware optimization improves accuracy, subgroup consistency, calibration, and generalizability.

Brain age has two fairness axes

One demographic axis and one target-distribution axis.

Race / ethnicity

NHW

Non-Hispanic
White

NHA

Non-Hispanic
African

HISP

Hispanic

Age-density

HiD

High Density
60-80

InD

Intermediate Density
50-60
80-90

LoD

Low Density
<50
>90

Race / ethnicity gap

MAE or BAG by NHW / NHA / HISP

Age-density gap

MAE or BAG by HiD / InD / LoD

Fairness here means lower subgroup error gaps, centered BAG, and stable calibration across groups.

Fairness-aware training improves accuracy and subgroup consistency

The selected Fair model jointly reduces overall error, race gap, and age-density gap.

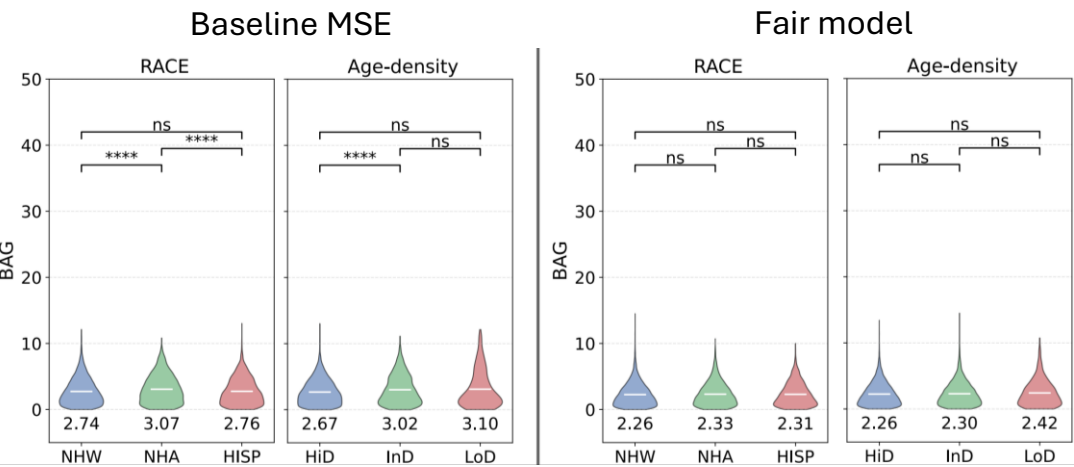
Model comparison

Metric	Baseline MSE	Fair model
Overall MAE	2.79 y	2.22 y
Race gap	0.22	0.09
Age-density gap	0.29	0.05
Fairness distance	0.58	0.03

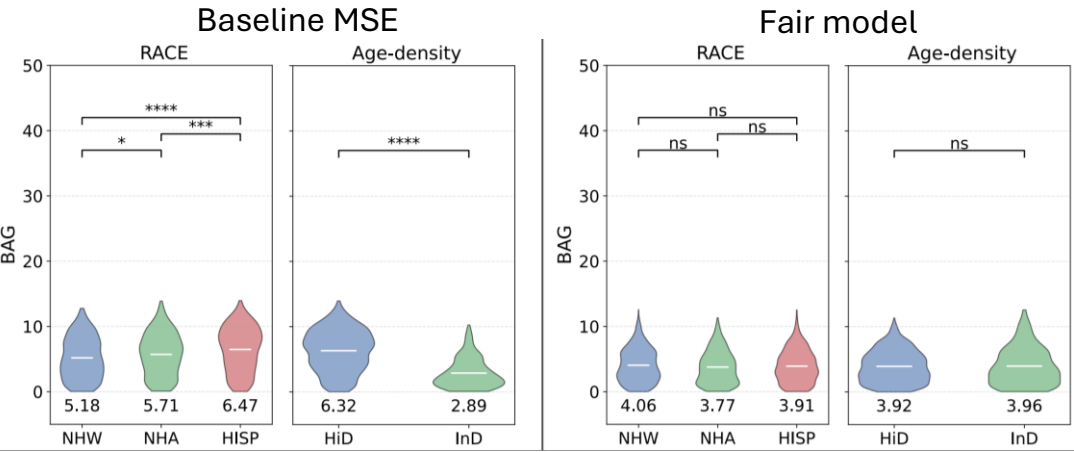
Conclusion

Fairness-aware optimization improved both global MAE and subgroup consistency in the latest model.

5-fold cross validation

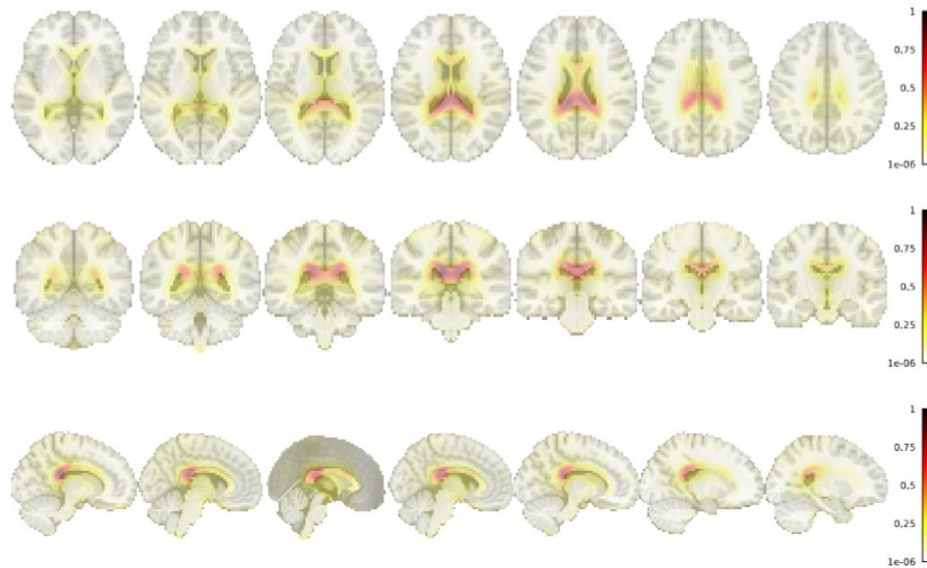


External DVCID set

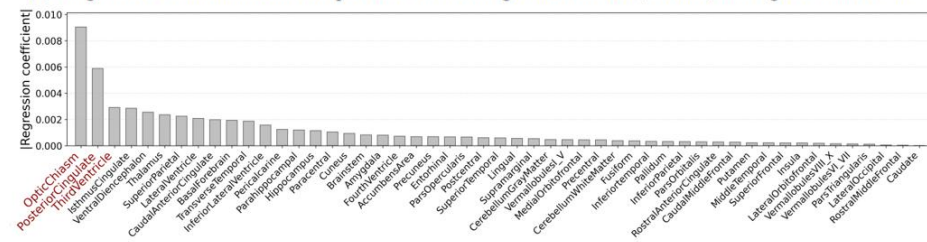


Fair model generalizes and improves calibration

Brain-age gap is useful only if errors are not systematically distorted.



(a) Integrated Gradients attribution maps of the Fair brain age model in axial, coronal, and sagittal orientations.



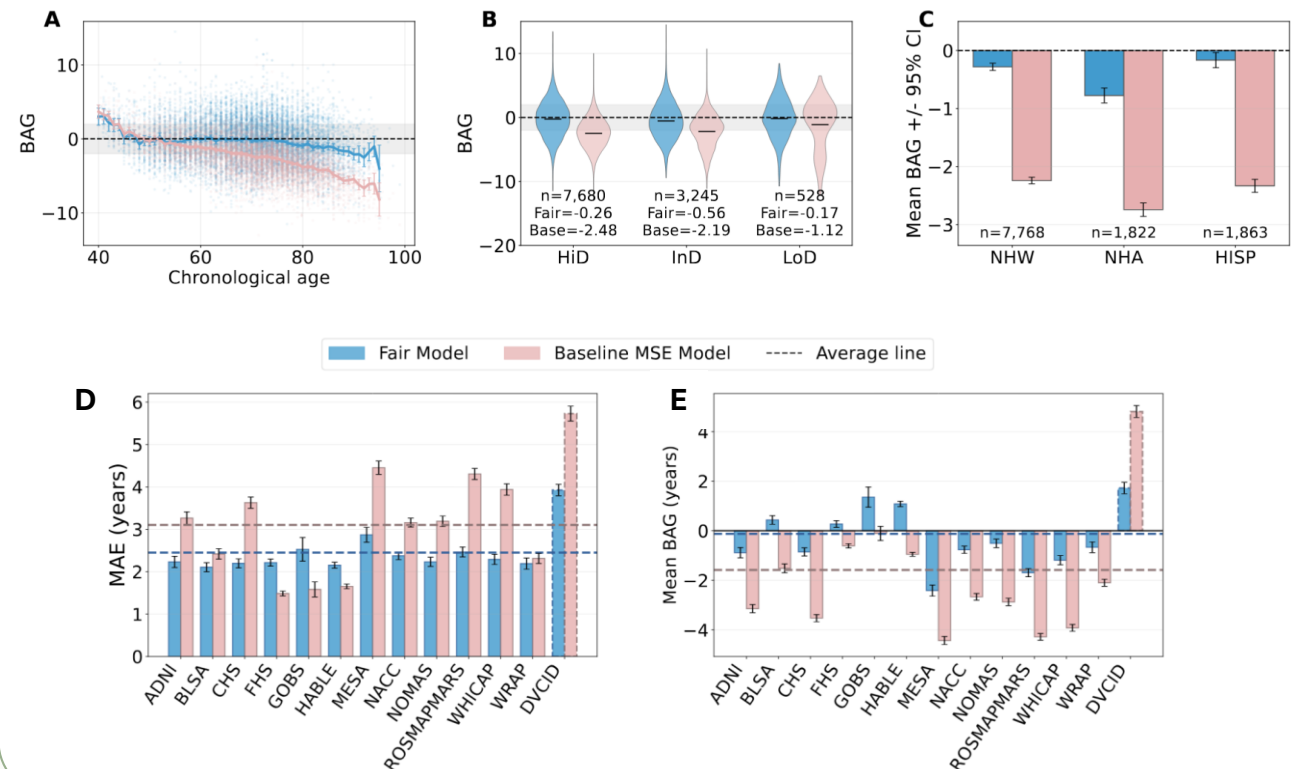
(b) Absolute linear regression coefficients between regional imaging features and predicted brain age.

Biological validity

IG maps emphasized plausible aging regions: ventricles, posterior cingulate, thalamus, medial temporal structures.

Interpretation

Fairness-aware brain age reduced systematic over-/under-estimation and produced more stable calibration across cohorts.



From fair diagnosis to fair biomarkers

Trustworthy neuroimaging AI across populations.

Dementia classification

- Disparities in FNR/FPR can persist even after balancing.
- Fairness-aware adaptation can reduce gaps.
- Explainability reveals population-specific model behavior.

Brain age prediction

- Fairness-aware training reduced MAE and subgroup gaps.
- External validation showed lower MAE and smaller BAG offsets.
- Calibration and biological validity support biomarker reliability.

Accuracy is necessary — but not sufficient — for trustworthy neuroimaging AI.

Thank you • Questions

Neuroimage Analytics Laboratory (NAL)

A collaborative home for neuroimaging AI, dementia research, and health equity.



Mohamad Habes



Nicolas Honnorat



David Martinez



Ngoc Huynh Ho



Niyas Shamsudeen
Kutty



Sokratis M Charisis



Di Wang



Mengjin Dong



Sachintha Ransara
Brandigampala



David Wang

Mission

Develop neuroimaging and machine-learning tools that are accurate, interpretable, and reliable across diverse populations.

Research focus

Brain aging and dementia biomarkers
Fairness and generalization across cohorts
Explainable AI for clinical translation
Collaborative cohort-based science